Republic of the Philippines
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GAZE: A Computer Vision-Based Head Movement Monitoring System for Classroom Examinations (CP2 Progress Report)

Project and completed work

GAZE helps instructors review visible head movements during examinations using one fixed camera. Its goal is to observe at least three students and record sustained or repeated movements. Head movement alone does not establish cheating. The current CPU prototype processes images and saved videos, marks faces and facial landmarks, assigns subject-relative forward, left, right or down labels, and records unclear observations as unknown.

Implemented features include shared image preparation, annotated image/video outputs, configurable seat areas, time histories, duration/repetition rules and local result records. A fresh-clone five-image run installed pinned dependencies and downloaded a checksum-verified model. All five faces received landmarks; one wrong direction and one unknown were retained. A further fresh-clone run reproduced the single-image command. These checks establish execution on the recorded desktop; independent member/laptop reproduction remains pending.

Prepared data and methods

The attributed Pointing04 subset contains 360 posed photographs from 15 people, all 384 x 288 pixels (Gourier et al., 2004). Validation uses 240 images from ten people (60 per class); the reserved test uses 120 images from five different people (30 per class). The 20 demonstration images are included within validation. Source pose annotations provide labels; file integrity, dimensions and representative pose directions were checked. CSV annotations, original JPEGs and the one-page data card accompany this report.

Training is not applicable: MediaPipe supplies pretrained landmarks (Google, 2026), and the project uses an untrained geometric direction rule. Prediction and evaluation share color conversion and proportional resizing; the selected small images retain their dimensions. An approximate OpenCV pose method provides a comparison (OpenCV, n.d.).

Initial measured results

Method	Correct / 240	Accuracy	Macro-F1
Initial geometric baseline	166	69.17%	0.6339
Validation-selected baseline	208	86.67%	0.8798
Approximate pose comparison	162	67.50%	0.6734

Source: saved GAZE validation measurements, October 2026. Unknown or missed observations count against accuracy. The selected baseline produced six unknowns. Its settings were tuned on this validation set, so 86.67% is a development result, not an independent final score. The reserved test remains unscored.

The public-domain FourPeople office proxy (Xiph.Org Foundation, n.d.) yielded 601 processed frames; at least three faces were detected in 547. A six-frame manual check matched 22 of 24 visible faces. This supports the saved-video demonstration, not classroom direction, seat or event accuracy. Six actual sample outputs and a gallery screenshot are submitted separately.

Effort-weighted completion: 50.0%

Completed effort / planned effort x 100 = 69 / 138 x 100 = 50.0%. The worksheet covers all 40 roadmap tasks, including evaluation, reports, demonstrations and three individual defenses. Eighteen acceptance-verified work tasks earn credit; two administrative tasks have zero weight. Unfinished tasks receive zero, including partly implemented work. Weights were estimated retrospectively as relative workload (1, 2, 3, 5 or 8 points), not logged hours, course marks or a previously agreed team baseline. The remaining effort is 69 points; team review of these assumptions is pending.

Blockers and contribution confirmation

Representative classroom footage and interval labels are missing; the posed-image subset does not cover natural writing, distant faces or occlusion. Historical desktop timing is not a current laptop benchmark. Implemented histories/rules still need classroom evaluation. Recorded implementation is under Jeremy Garin with coding-agent assistance; Marc Justin Alberto and Jose III Aquino must confirm their own contributions and explain their code. The 50.0% estimate does not certify checkpoint acceptance. Revised work targets require team confirmation; course deadlines remain unchanged.

References

Google. (2026, October 9). *Face landmark detection guide*. Google AI Edge. https://developers.google.com/edge/mediapipe/solutions/vision/face_landmarker

Gourier, N., Hall, D., & Crowley, J. L. (2004). *Estimating face orientation from robust detection of salient facial structures* [Workshop paper]. Pointing 2004, Cambridge, United Kingdom. https://homepages.inf.ed.ac.uk/rbf/CAVIAR/PAPERS/Pointing04-Gourier.pdf

OpenCV. (n.d.). *Perspective-n-Point (PnP) pose computation*. https://docs.opencv.org/4.11.0/d5/d1f/calib3d_solvePnP.html

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